Minh-Duc VU

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Summary — AI Research Engineer with research experience in Spiking Neural Networks, deep learning, and remote sensing, focusing on efficient model optimization under resource constraints. Skilled in developing reproducible benchmarks, designing efficient models, and contributing to peer-reviewed publications. Proven ability to bridge innovative research with real-world deployment in both academic and industrial environments.

Skills

Programming Python, JavaScript, Solidity
Databases MongoDB, SQL
Frameworks PyTorch, TensorFlow, Streamlit
Tools Apache Airflow, Git, Trello
Language Vietnamese, English, French

Experience

L2TI Laboratory - LabCom IRISER (ANR-21-LCV3-0004) **Oct 2024 – Present, France**
Research & Development Engineer

Benchmark Dataset CADOT (**C**ityscape **A**erial Image **D**ataset for **O**bject De**T**ection)

- Designed and implemented the CADOT benchmark for realistic remote sensing object detection.
- Created a semi-automatic annotation workflow to streamline dataset labeling.
- Evaluated the dataset using baseline models for performance and efficiency.
- Developed and maintained the official dataset website: cadot.onrender.com/dataset

IEEE ICIP 2025 Dataset Paper

- Co-authored a dataset paper accepted to IEEE ICIP 2025, introducing CADOT and its evaluation framework.
- Documented key processes and system configurations to support collaboration and reproducibility.

IEEE ICIP 2025 Grand Challenge

- Co-organizer of the ICIP 2025 Grand Challenge on remote sensing object detection.
- Designed the evaluation protocols and curated competition datasets.
- Developed and deployed the challenge website: https://cadot.onrender.com
- Coordinated infrastructure setup and provided technical support during challenge rollout.

Research on Spiking Neural Networks

- Conducted research on Spiking Neural Networks (SNNs) focusing on energy-efficient inference for embedded systems.
- Implemented SNN models for object detection tasks, benchmarking accuracy–latency–energy trade-offs on constrained hardware.
- Explored optimization strategies (e.g., pruning, quantization, event-driven processing) to improve inference speed under real-time constraints.

TensorRT Optimization

- Built and optimized a TensorRT inference pipeline for YOLO models on NVIDIA GPUs
- Reduced inference latency by 0.7 ms per image with only a 0.03 accuracy trade-off
- Improved training and deployment efficiency through model and workflow optimization

Cross-view Geo-localization

- Enhanced the CADOT dataset by integrating GPS metadata for each image and object annotation

L2TI Laboratory - Multimedia Team **Mar 2024 – Sep 2024, France**
Research Intern

- Investigated Masked Image Modeling (MIM) techniques to enhance feature learning in remote sensing imagery using self-supervised approaches.
- Proposed a novel architecture combining Swin Transformer backbone with YOLOv5 detection head to improve multi-scale feature fusion.
- Improved SimMIM performance by implementing a Cross-Attention Decoder, boosting pretraining quality for downstream detection tasks.
- Co-authored a scientific paper on arXiv, exploring MIM-based pretraining for multi-modal satellite imagery.

ChainOS Solution **Mar 2021 – Feb 2023, Vietnam**
Blockchain Developer

- Developed smart contracts and backend systems using Solidity and NodeJS for GameFi and NFT platforms.
- Contributed to projects including:
  - **Kanimals Clash (03.2022 – 02.2023)** – Team Lead; developed smart contracts, managed team, and provided backend and web support. Stack: Solidity, NodeJS, PHP, Truffle, ReactJS, ExpressJS.
  - **Hydraverse (12.2021 – 12.2022)** – Built swap token contracts, signing system, and staking/vesting site. Stack: Solidity, NodeJS, MongoDB Compass.

  - **12 Guardians (01.2022 – 03.2022)** – Created smart contracts and backend web services for blockchain interactions.
  - **NFT Gold (06.2021 – 09.2021)** – PHP Web Developer; responsible for web support using Laravel.
- Experience with tools such as Truffle, ReactJS, ExpressJS, Laravel, and MongoDB Compass.
- Mentored junior developers and handled incident resolution in live smart contract deployments.

G-Group **Jun 2020 – Dec 2020, Vietnam**
Computer Vision Engineer Intern

- Researched neural architectures (SSD, YOLOv3/v5, Darknet, Inception, VGG-16) for object detection.
- Implemented models for real-world computer vision tasks such as detection and tracking.
- Applied academic theory to production-level CV projects, improving practical expertise.

KSE Solution **Nov 2019 – Mar 2020, Vietnam**
Computer Vision Engineer Intern

- Explored image processing techniques and implemented object detection/classification pipelines.
- Gained hands-on experience with TensorFlow and PyTorch for CV tasks.
- Contributed to team-driven experimentation and model development workflows.

Education

EPITA **Feb 2023 – Sep 2024, France**
Master of Science in Computer Science
Majors: Data Science and Analytics

FPT APTECH **Apr 2018 – June 2021, Vietnam**
Higher Diploma in Software Engineering
Majors: Software Engineering

Hanoi University of Science **Sep 2017 – Dec 2021, Vietnam**
Bachelor of Computer and Information Science
Majors: Computer and Information Science

Publications

- **Minh Duc Vu**, Anissa Mokraoui, Fangchen Feng, Bouzid Arezki, Borel Sonna, Bissmella Bahaduri, Hicham Talaoubrid. *CADOT: Cityscape Aerial Image Dataset for Object Detection*. In: 2025 IEEE International Conference on Image Processing (ICIP), pp. 2636–2641, 2025. DOI: 10.1109/ICIP55913.2025.11084552.

- **Minh Duc Vu**, Zuheng Ming, Fangchen Feng, Bissmella Bahaduri, Anissa Mokraoui. *Interactive Masked Image Modeling for Multimodal Object Detection in Remote Sensing*. arXiv preprint arXiv:2409.08885, September 2024.